Practice workbook · Draw or label the parts of a synovial joint

Internal structure of a synovial joint shown as a diagram.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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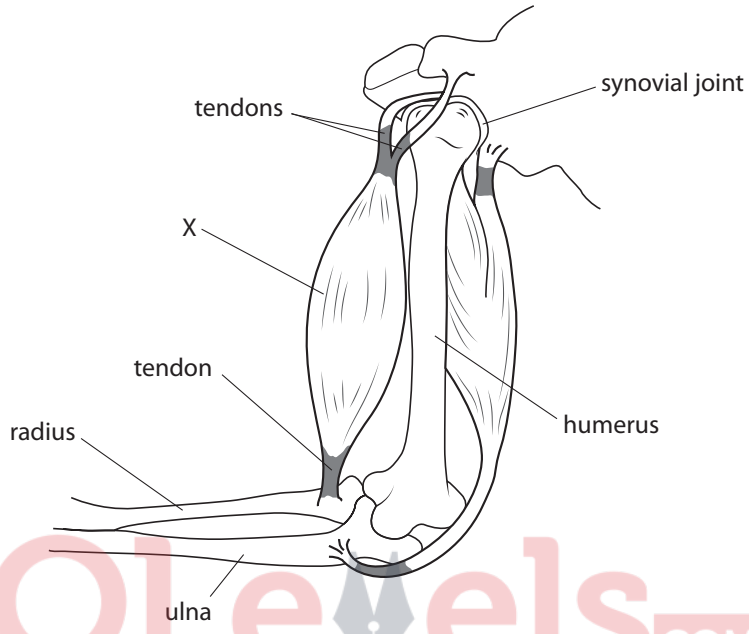
Type ID: HBI-SYNOVIAL-JOINT-STRUCTURE

1. Question 7 · part b · 6 marks

November 2020 Paper 1

Worked-example retry · Attempt part b only.

7 The diagram shows the bones of a human arm and the muscles that move the lower arm.



(a) (i) Muscle X needs energy to contract.

Name the process that supplies this energy.

(1)

(ii) Muscle X contracts when the lower arm is raised further.

Describe the changes in shape of muscle X when the lower arm is raised further.

(2)

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1. Question 7 · part b · 6 marks

November 2020 Paper 1

Worked-example retry · Attempt part b only.

(iii) Explain how the contraction of muscle X causes the lower arm to be raised.

(3)

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(b) Draw a labelled diagram to show the internal structure of the synovial joint.

(6)

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(Total for Question 7 = 12 marks)

1. Question 7 · part b · 6 marks · Working and answer

Working and answer

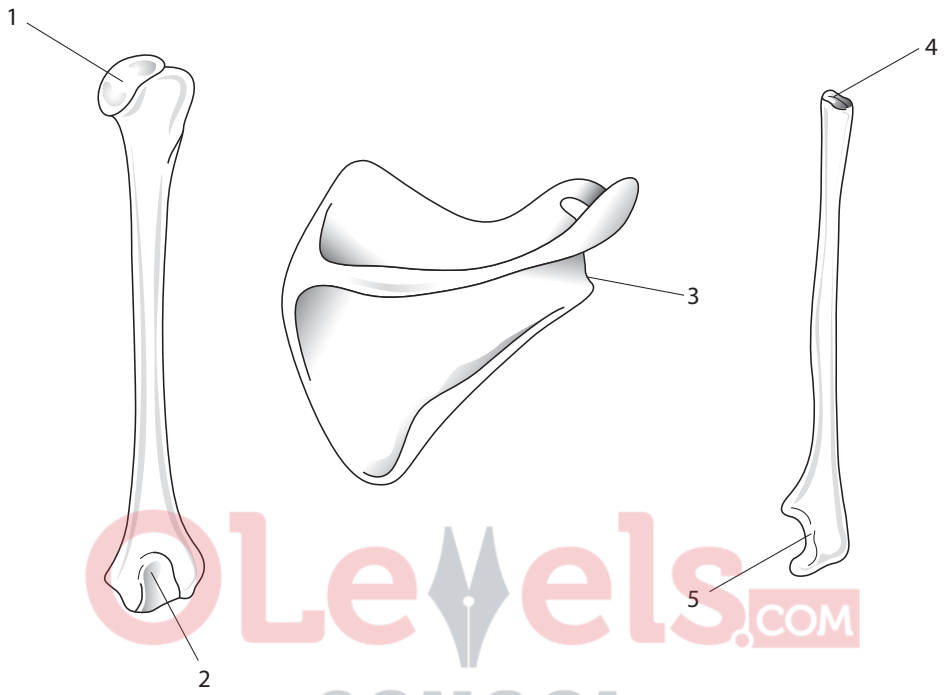
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2. Question 2 · part b.iii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.iii only.

2 (a) The diagrams show three bones from the human skeleton.



Which two numbered regions come together to form a ball and socket joint?

(1)

- **A** 1 and 3
- **B** 1 and 5
- **C** 2 and 3
- **D** 2 and 4

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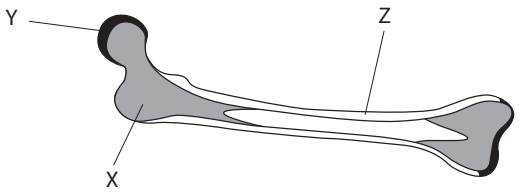
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2. Question 2 · part b.iii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.iii only.

(b) The diagram shows a section through a bone.



(i) Identify the part labelled X.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

(1)

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(iv) Describe the function of the part labelled Y.

(2)

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2. Question 2 · part b.iii · 1 mark · Working and answer

Working and answer

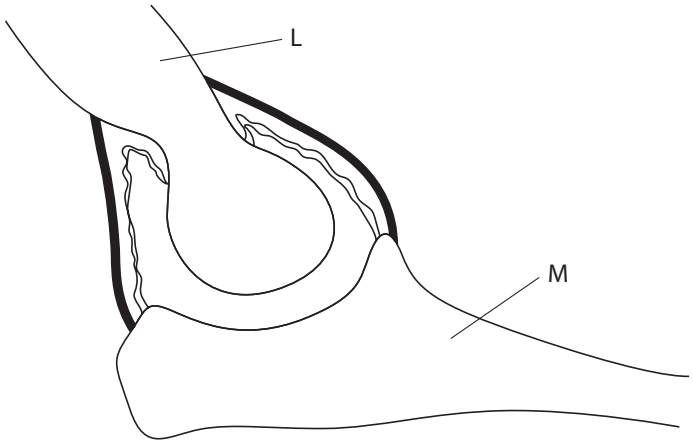
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3. Question 6 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

6 The diagram shows some of the structures of the elbow joint.



(a) State what is meant by the term **joint**.

(2)

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(b) (i) Give the name of bone L and the name of bone M.

(2)

L .....

M .....

(ii) Cartilage has not been shown on the diagram.

Draw the missing cartilage on the diagram.

(2)

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3. Question 6 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

(c) Explain the effects on a person if the cartilage in this joint is missing or damaged.

(2)

(d) Tendons are important in the functioning of a joint.

Explain the importance of tendons in the movement of bone M.

(4)

(Total for Question 6 = 12 marks)

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3. Question 6 · part b.ii · 2 marks · Working and answer

Working and answer

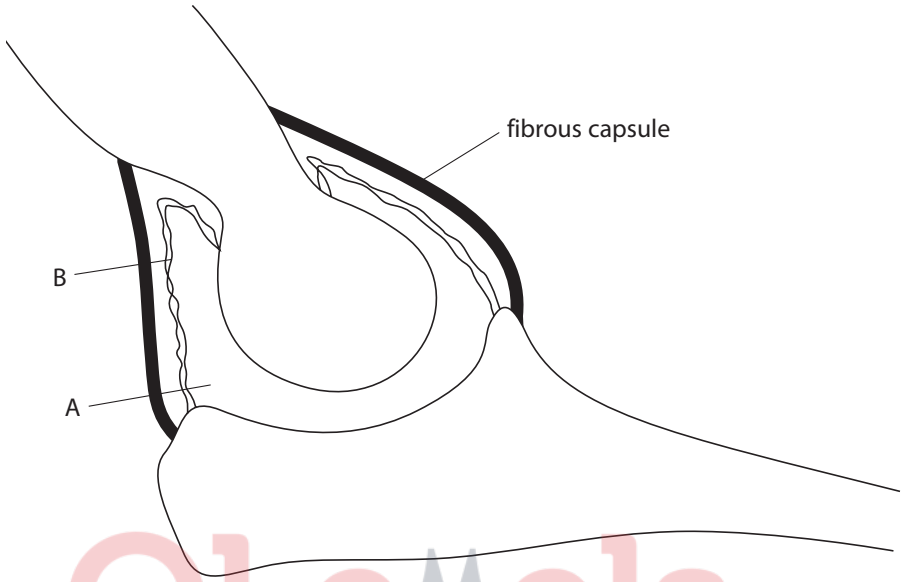
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4. Question 3 · part a.i · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.i only.

3 The diagram shows part of an elbow joint without any cartilage.



(a) (i) Identify part A and part B. **(2)**

part A .....

part B .....

(ii) Describe the function of part A. **(2)**

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(b) (i) Complete the diagram by drawing the cartilage. **(2)**

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4. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

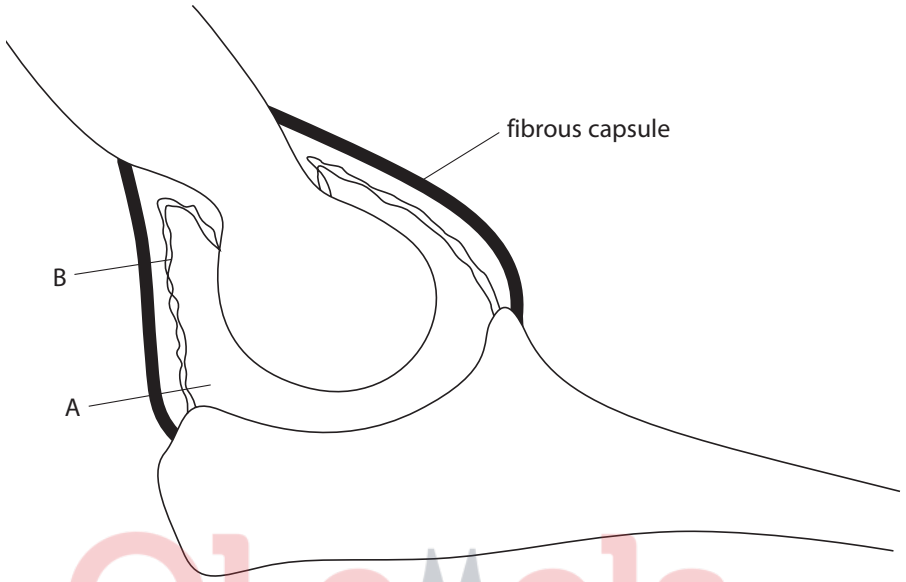
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5. Question 3 · part b.i · 2 marks

January 2022 Paper 1

Further practice · Attempt part b.i only.

3 The diagram shows part of an elbow joint without any cartilage.



(a) (i) Identify part A and part B. **(2)**

part A .....

part B .....

(ii) Describe the function of part A. **(2)**

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(b) (i) Complete the diagram by drawing the cartilage. **(2)**

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5. Question 3 · part b.i · 2 marks

January 2022 Paper 1

Further practice · Attempt part b.i only.

(ii) Cartilage often breaks down as people become older.

Explain the effects this has on the working of the joint.

(2)

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(c) Explain the role of tendons in the movement of the lower arm.

(5)

(Total for Question 3 = 13 marks)

5. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

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